LECTURE-6

Cell

Acknowledgements: Some of the materials are taken from v different sources in Google besides from Campbell's book on biology

INDRAPRASTHA INSTITUTE *of*
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DELHI

Overview

KEY CONCEPTS

6.1 Biologists use microscopes and the tools of biochemistry to study cells

6.2 Eukaryotic cells have internal membranes that compartmentalize their functions

6.3 The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes

6.4 The endomembrane system regulates protein traffic and performs metabolic functions in the cell

6.5 Mitochondria and chloroplasts change energy from one form to another

6.6 The cytoskeleton is a network of fibers that organizes structures and activities in the cell

6.7 Extracellular components and connections between cells help coordinate cellular activities

Cells -- I am classified !!!

- From the evolutionary point of view, cells have been, traditionally, divided into two categories, viz., prokaryotes and eukaryotes.

- **Prokaryotes**: Most of the common bacteria like, for example, Escherichia Coli (E. coli) and Salmonella.

- **Eukaryotes**: Animals, plants and fungi are collectively called eukaryotes.

- **Major difference** between prokaryotic and eukaryotic cells lies in their internal architectures; the main distinct feature of eukaryotic cells is the cell nucleus where the genetic materials are stored.

Properties of cells

❖ Cell is dynamic as it involves in various processes.

❖ Cell can

✓ change its shape, i.e., it is elastic

✓ communicate with other cells (cell-to-cell communication)

✓ move (molecular motors)

✓ process signals and involves in transport of molecules (Signalling process)

✓ divide (Mitotic) and can make multiple copies (Replication)

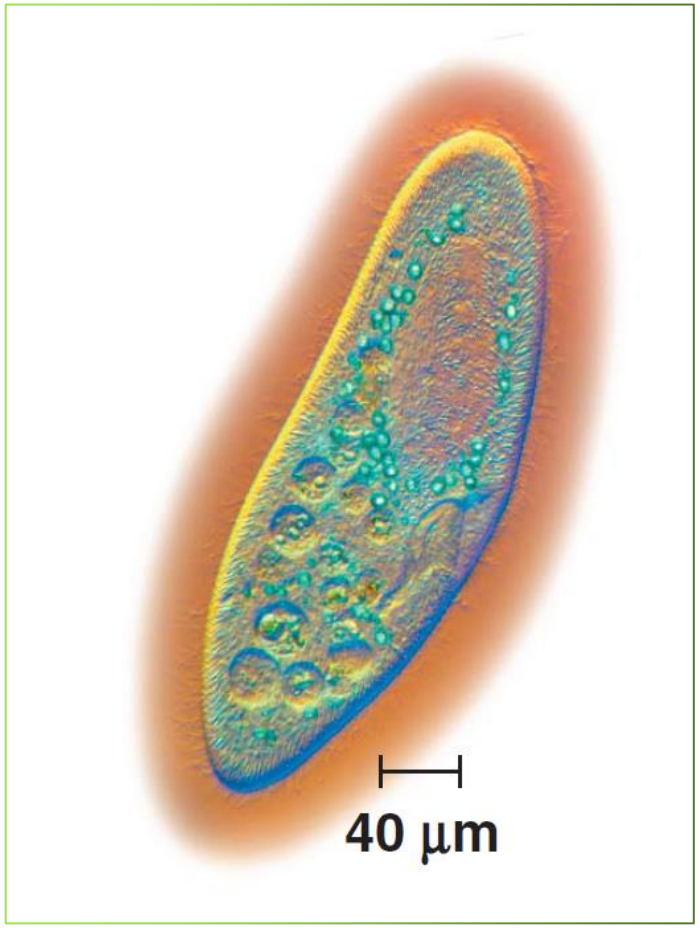
Cell: The fundamental unit of life

- All organisms are made of cells.
  - Many organisms are single-celled.

- **The cell is the simplest collection of matter that can be alive.**

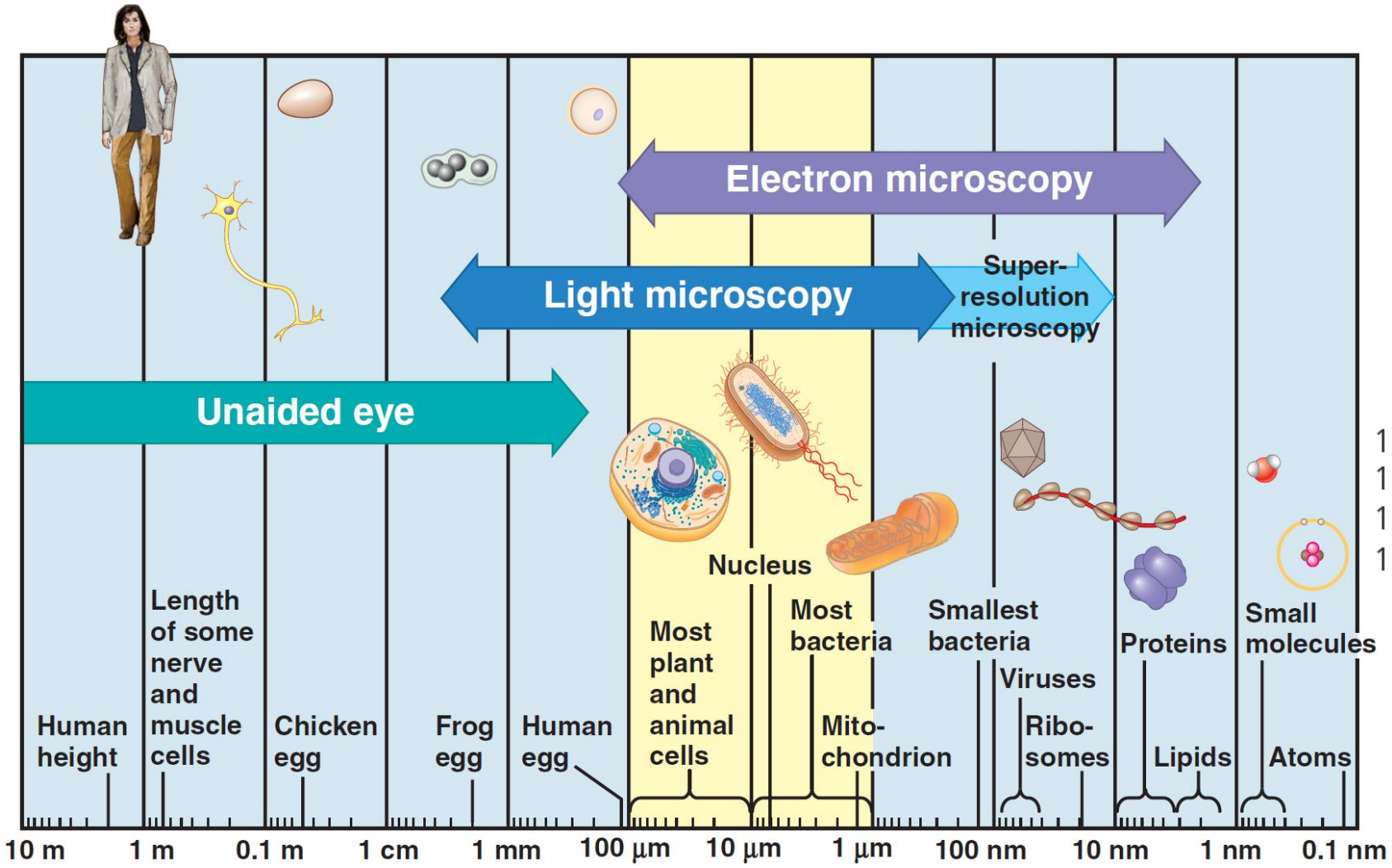
- Even when arranged into higher levels of organization, such as tissues and organs, **cells are an organism's basic units of structure and function**.

- Evolution is the unifying biological theme; all cells are related by their descent from earlier cells but have been modified in various ways during the history of life on Earth.



Paramecium, a single-celled organism

Tools to see and understand cells



The discovery and early study of cells progressed with the invention of microscopes in 1590 and their improvement in the 17th century.

Conversion

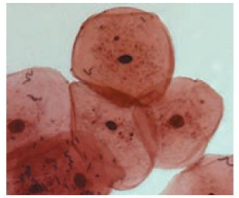
1 centimeter (cm) = 10^{-2} meter (m) = 0.4 inch
1 millimeter (mm) = 10^{-3} m
1 micrometer (µm) = 10^{-3} mm = 10^{-6} m
1 nanometer (nm) = 10^{-3} µm = 10^{-9} m

MICROSCOPY

Light Microscopy (LM)

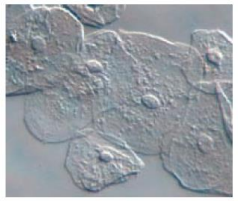
Brightfield (unstained specimen). Light passes directly through the specimen. Unless the cell is naturally pigmented or artificially stained, the image has little contrast. (The first four light micrographs show human cheek epithelial cells; the scale bar pertains to all four micrographs.)

Brightfield (stained specimen). Staining with various dyes enhances contrast. Most staining procedures require that cells be fixed (preserved), thereby killing them.

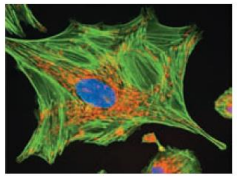


Phase-contrast. Variations in density within the specimen are amplified to enhance contrast in unstained cells; this is especially useful for examining living, unpigmented cells.

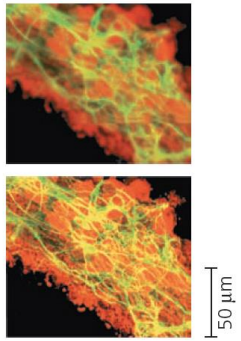
Differential-interference-contrast (Nomarski). As in phase-contrast microscopy, optical modifications are used to exaggerate differences in density; the image appears almost 3-D.



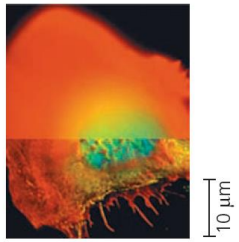
Fluorescence. The locations of specific molecules in the cell can be revealed by labeling the molecules with fluorescent dyes or antibodies; some cells have molecules that fluoresce on their own. Fluorescent substances absorb ultraviolet radiation and emit visible light. In this fluorescently labeled uterine cell, nuclear material is blue, organelles called mitochondria are orange, and the cell's "skeleton" is green.



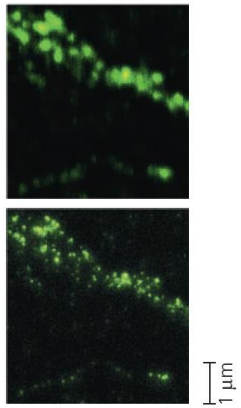
Confocal. The top image is a standard fluorescence micrograph of fluorescently labeled nervous tissue (nerve cells are green, support cells are orange, and regions of overlap are yellow); below it is a confocal image of the same tissue. Using a laser, this "optical sectioning" technique eliminates out-of-focus light from a thick sample, creating a single plane of fluorescence in the image. By capturing sharp images at many different planes, a 3-D reconstruction can be created. The standard image is blurry because out-of-focus light is not excluded.



Deconvolution. The top of this split image is a compilation of standard fluorescence micrographs through the depth of a white blood cell. Below is an image of the same cell reconstructed from many blurry images at different planes, each of which was processed using deconvolution software. This process digitally removes out-of-focus light and reassigns it to its source, creating a much sharper 3-D image.



Super-resolution. On the top is a confocal image of part of a nerve cell, using a fluorescent label that binds to a molecule clustered in small sacs in the cell (vesicles) that are 40 nm in diameter. The greenish-yellow spots are blurry because 40 nm is below the 200-nm limit of resolution for standard light microscopy. Below is an image of the same part of the cell, seen using a new super-resolution technique. Sophisticated equipment is used to light up individual fluorescent molecules and record their position. Combining information from many molecules in different places "breaks" the limit of resolution, resulting in the sharp greenish-yellow dots seen here. (Each dot is a 40-nm vesicle.)



- Although an LM can resolve individual cells, it cannot resolve much of the internal anatomy, especially the **organelles** and membrane-enclosed structures within eukaryotic cells.

- To resolve smaller structures, scientists use an **electron microscope (EM)**, which focuses a beam of electrons through the specimen or onto its surface.

  - Because the resolution is inversely related to the wavelength used for imaging, EMs (whose electron beams have shorter wavelengths than visible light) have finer resolution than LMs.
  - Theoretically, the resolution of a modern EM could reach 0.002 nm, but the practical limit is closer to about 2 nm.

Microscopy

Microscopes and the tools of biochemistry for the study of cells

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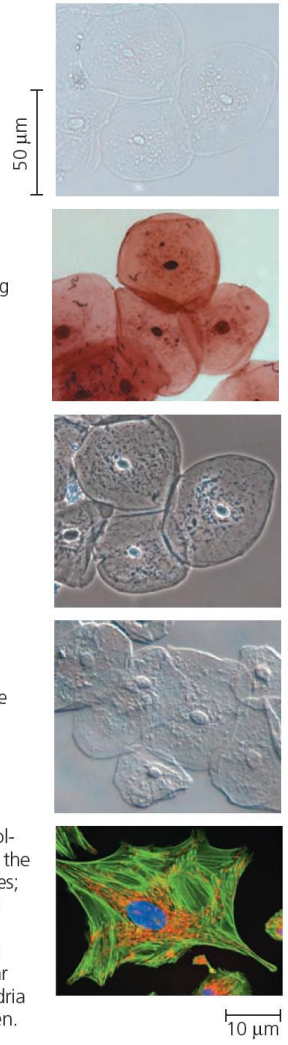

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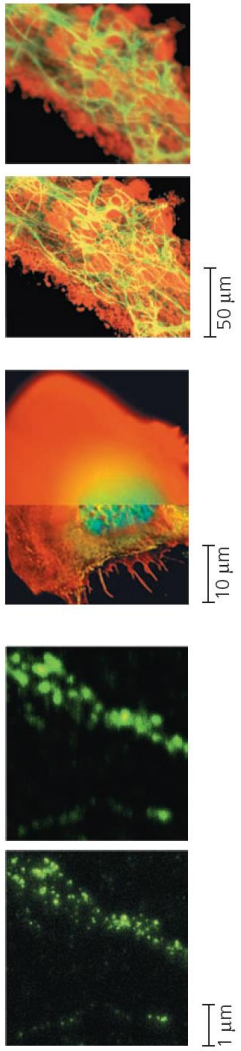


- **Scanning electron microscopes (SEMs)** are useful for studying a specimen's surface structure or topography.
  - The sample surface is covered with a thin film of gold.
  - The beam excites electrons on the sample's surface, and these secondary electrons are collected and focused on a screen, producing a surface image of the specimen.
  - SEMs have a great depth of field, resulting in an image that seems three-dimensional.

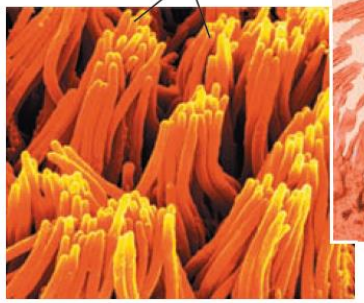
Microscopes and the tools of biochemistry for the study of cells

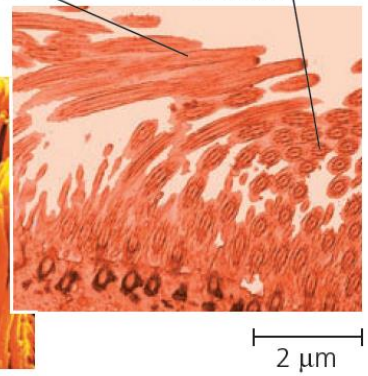
- **Transmission electron microscopes (TEMs)** are used to study the internal structure of cells.
  - A TEM aims an electron beam through a very thin section of the specimen.
  - To enhance contrast, the thin sections are stained with atoms of heavy metals, which attach to certain cellular structures.
  - The image is focused and magnified by electromagnets.

Electron Microscopy (EM)

Scanning electron microscopy (SEM). Micrographs taken with a scanning electron microscope show a 3-D image of the surface of a specimen. This SEM shows the surface of a cell from a trachea (windpipe) covered with cilia. Beating of the cilia helps move inhaled debris upward toward the throat. Electron micrographs are black and white, but are often artificially colorized to highlight particular structures, as has been done with both micrographs (SEM and TEM) shown here.

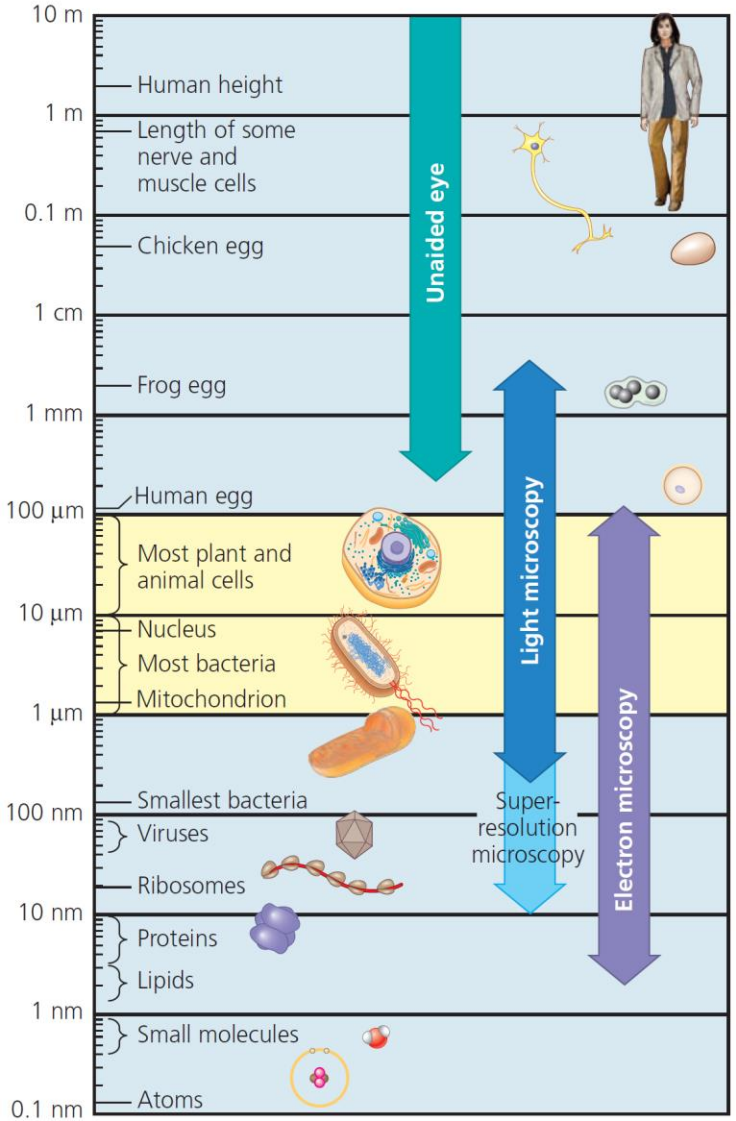
Abbreviations used in figure legends in this book:
LM = Light Micrograph
SEM = Scanning Electron Micrograph
TEM = Transmission Electron Micrograph





Transmission electron microscopy (TEM). A transmission electron microscope profiles a thin section of a specimen. Here we see a section through a tracheal cell, revealing its internal structure. In preparing the specimen, some cilia were cut along their lengths, creating longitudinal sections, while other cilia were cut straight across, creating cross sections.

Tools to study cells



- Microscopes vary in magnification, resolution, and contrast.
- ***Magnification*** is the ratio of an object's image to its real size. A light microscope can magnify effectively to about 1,000 times the real size of a specimen.
- ***Resolution*** is a measure of image clarity. It is the minimum distance two points can be separated and still be distinguished as two separate points. The minimum resolution of an LM is about 200 nanometers (nm), the size of a small bacterium.
- ***Contrast*** accentuates differences in parts of the sample. It can be improved by staining or labeling of cell components so they stand out.

Electron Microscopy (EM)

Scanning electron microscopy (SEM). Micrographs taken with a scanning electron microscope show a 3-D image of the surface of a specimen. This SEM shows the surface of a cell from a trachea (windpipe) covered with cilia. Beating of the cilia helps move inhaled debris upward toward the throat. Electron micrographs are black and white, but are often artificially colorized to highlight particular structures, as has been done with both micrographs (SEM and TEM) shown here.

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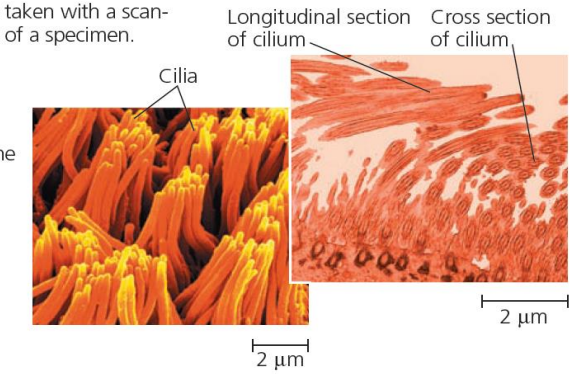

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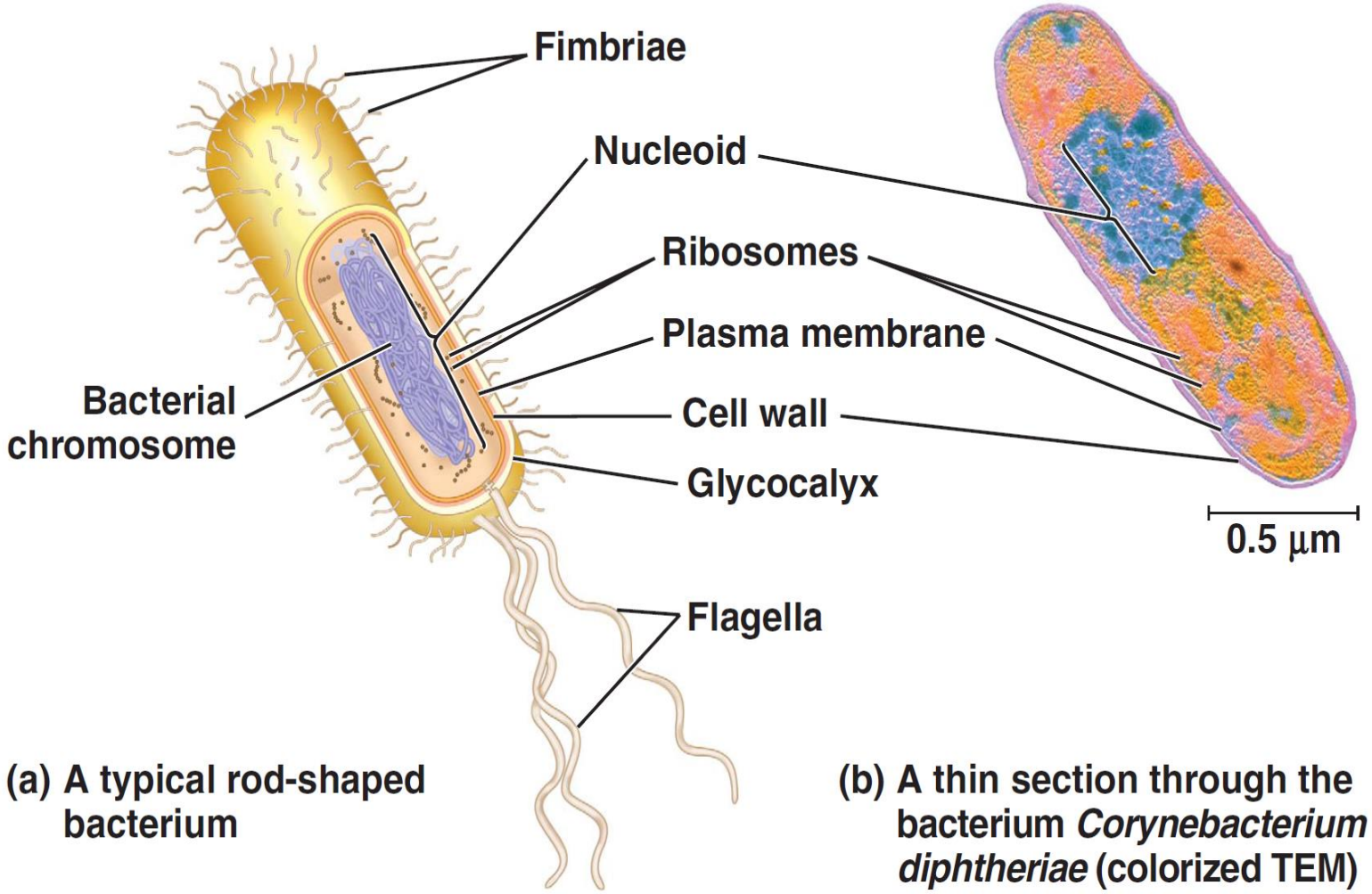
Eukaryotic cells



(a) A typical rod-shaped bacterium

(b) A thin section through the bacterium *Corynebacterium diphtheriae* (colorized TEM)

- All cells are surrounded by a selective barrier, the *plasma membrane*.
- The semifluid substance within the membrane is the **cytosol**, in which subcellular components are suspended.
- A major difference between prokaryotic and eukaryotic cells is the location of the DNA.

Eukaryotic cells: internal membrane for functional segregation

- In a eukaryotic cell, most of the DNA is in an organelle bounded by a double membrane, the *nucleus*.
- In a prokaryotic cell, the DNA is concentrated in the *nucleoid*, without a membrane separating it from the rest of the cell.

- The interior of a prokaryotic cell and the region between the nucleus and the plasma membrane of a eukaryotic cell is the *cytoplasm*.
- Within the cytoplasm of a eukaryotic cell are a variety of membrane-bound organelles with specialized form and function. These *membrane-bound organelles are absent in prokaryotes*.

- Eukaryotic cells are generally much larger than prokaryotic cells.

- The logistics of carrying out cellular metabolism set limits on cell size.

  - *At the lower limit, the smallest bacteria, mycoplasmas, are 0.1–1.0 µm in diameter.*
  - *Most bacteria are 1–5 µm in diameter.*
  - *Eukaryotic cells are typically 10–100 µm in diameter*

Structures based on their functions

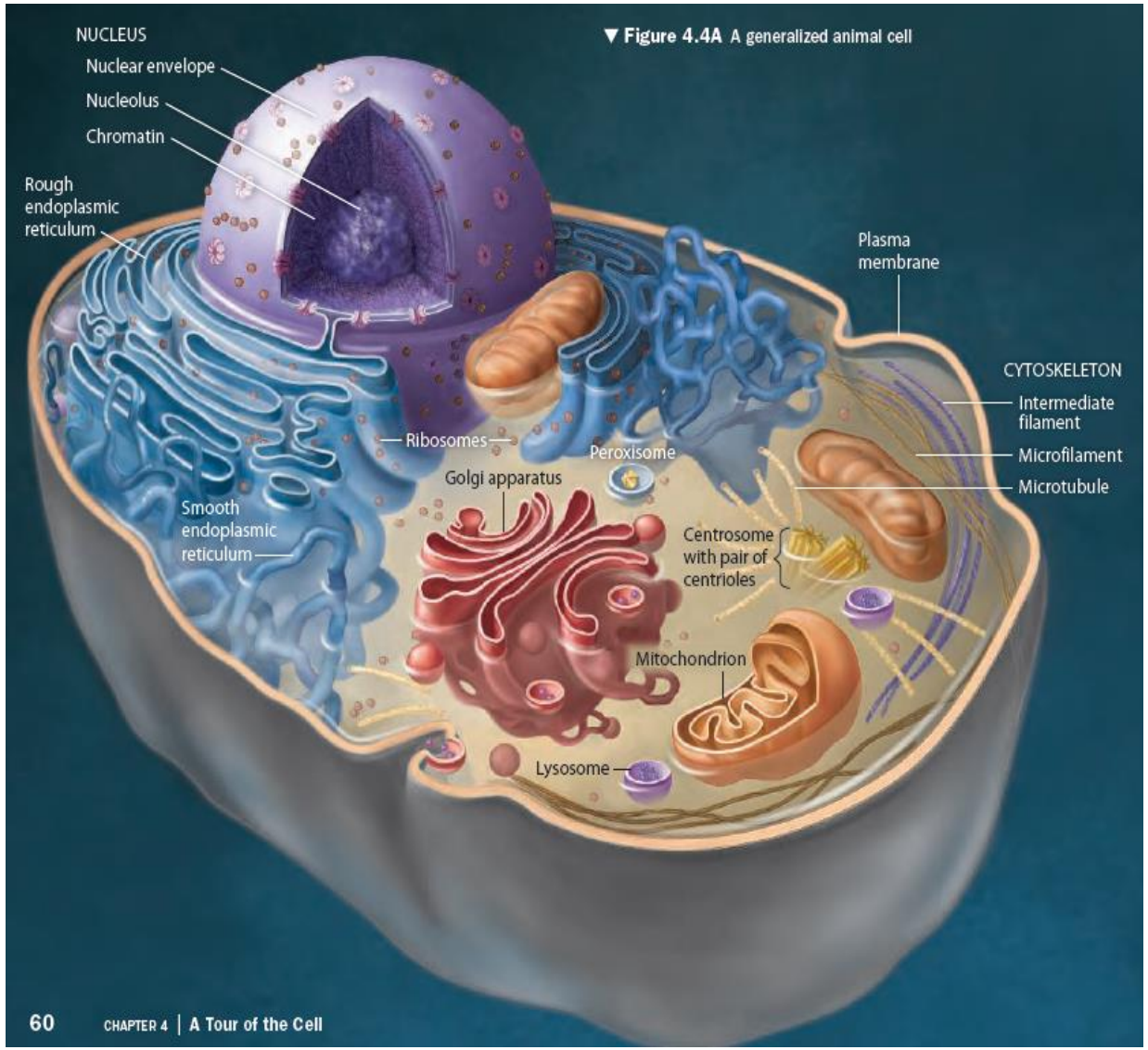
The organelles and other structures of eukaryotic cells can be organized into four basic functional groups:

1. The nucleus and ribosomes carry out the genetic control of the cell.

2. Organelles involved in the manufacture, distribution, and breakdown of molecules include the endoplasmic reticulum, Golgi apparatus, lysosomes, vacuoles, and peroxisomes.

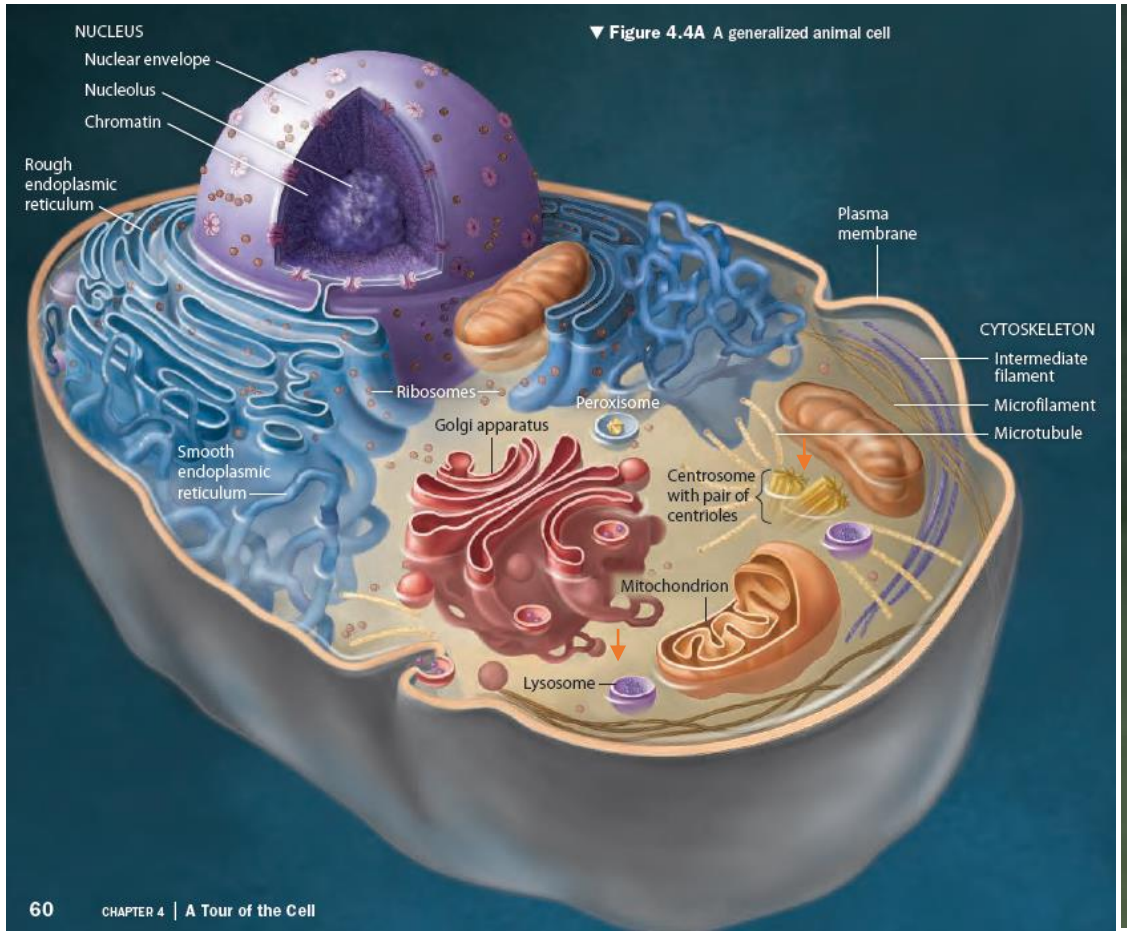
3. Mitochondria in all cells and chloroplasts in plant cells function in energy processing.

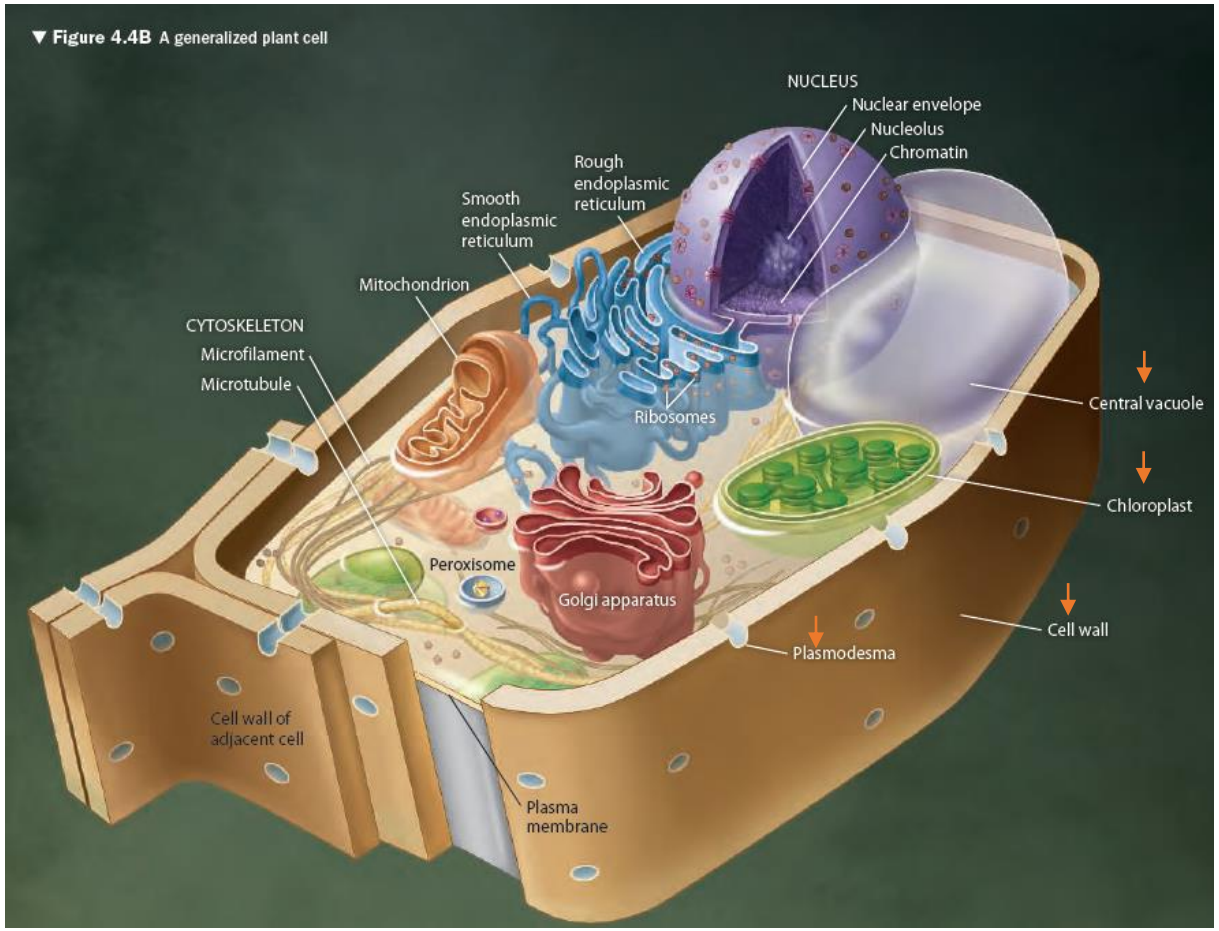
4. Structural support, movement and communication between cells are the functions of the cytoskeleton, plasma membrane, and plant cell wall.



▼ Figure 4.4A A generalized animal cell

Animal and plant cell–a comparison





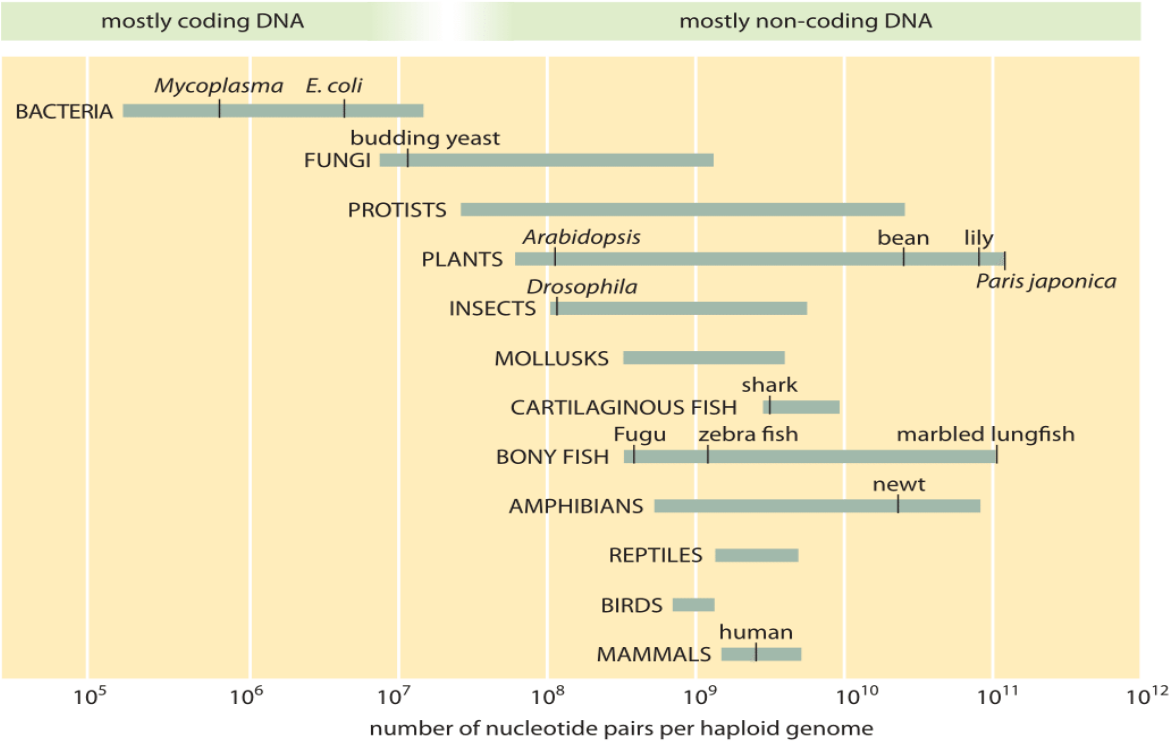
1. Identify the plant cell structures not present in the animal cell.

2. Identify the animal cell structures not present in the plant cell.

DNA size of various species

- Metabolic requirements also set an upper limit to the size of a single cell.

- The *plasma membrane* functions as a selective barrier that allows the passage of oxygen, nutrients, and wastes for the whole volume of the cell.



Organism	Genome size* (nucleotide pairs)	Approximate number of genes
Homo sapiens (human)	3200×10^6	30,000
Mus musculus (mouse)	2800×10^6	30,000
Drosophila melanogaster (fruit fly)	200×10^6	15,000
Arabidopsis thaliana (plant)	220×10^6	29,000
Caenorhabditis elegans (roundworm)	130×10^6	21,000
Saccharomyces cerevisiae (yeast)	13×10^6	6600
Escherichia coli (bacteria)	4.6×10^6	4300

1 bp = one base pairs;
1 M = one Mega = 10^6
1 G = one Giga = 10^9.

About human genome

- The nuclear genome comprises approximately 3 2000 000 000 nucleotides (32Gbp) of DNA, divided into 24 linear molecules (chromosomes), the shortest 50 000 000 nucleotides in length and the longest 260 000 000 nucleotides, each contained in a different chromosome.

- These 24 chromosomes consist of 22 autosomes and the two sex chromosomes, X and Y.

- The mitochondrial genome is a circular DNA molecule of 16 569 nucleotides, multiple copies of which are located in the energy-generating organelles called mitochondria.

- Each of the approximately 1013 cells in the adult human body has its own copy or copies of the genome, the only exceptions being those few cell types, such as red blood cells, that lack a nucleus in their fully differentiated state.

- The vast majority of cells are diploid and so have two copies of each autosome, plus two sex chromosomes, XX for females or XY for males - 46 chromosomes in all.

- These are called somatic cells, in contrast to sex cells or gametes, which are haploid and have just 23 chromosomes, comprising one of each autosome and one sex chromosome. Both types of cell have about 8000 copies of the mitochondrial genome, 10 or so in each mitochondrion.

Specific parts of cell: Nucleus and DNA

- The **nucleus** contains most of the genes in a eukaryotic cell. Additional genes are located in mitochondria and chloroplasts.

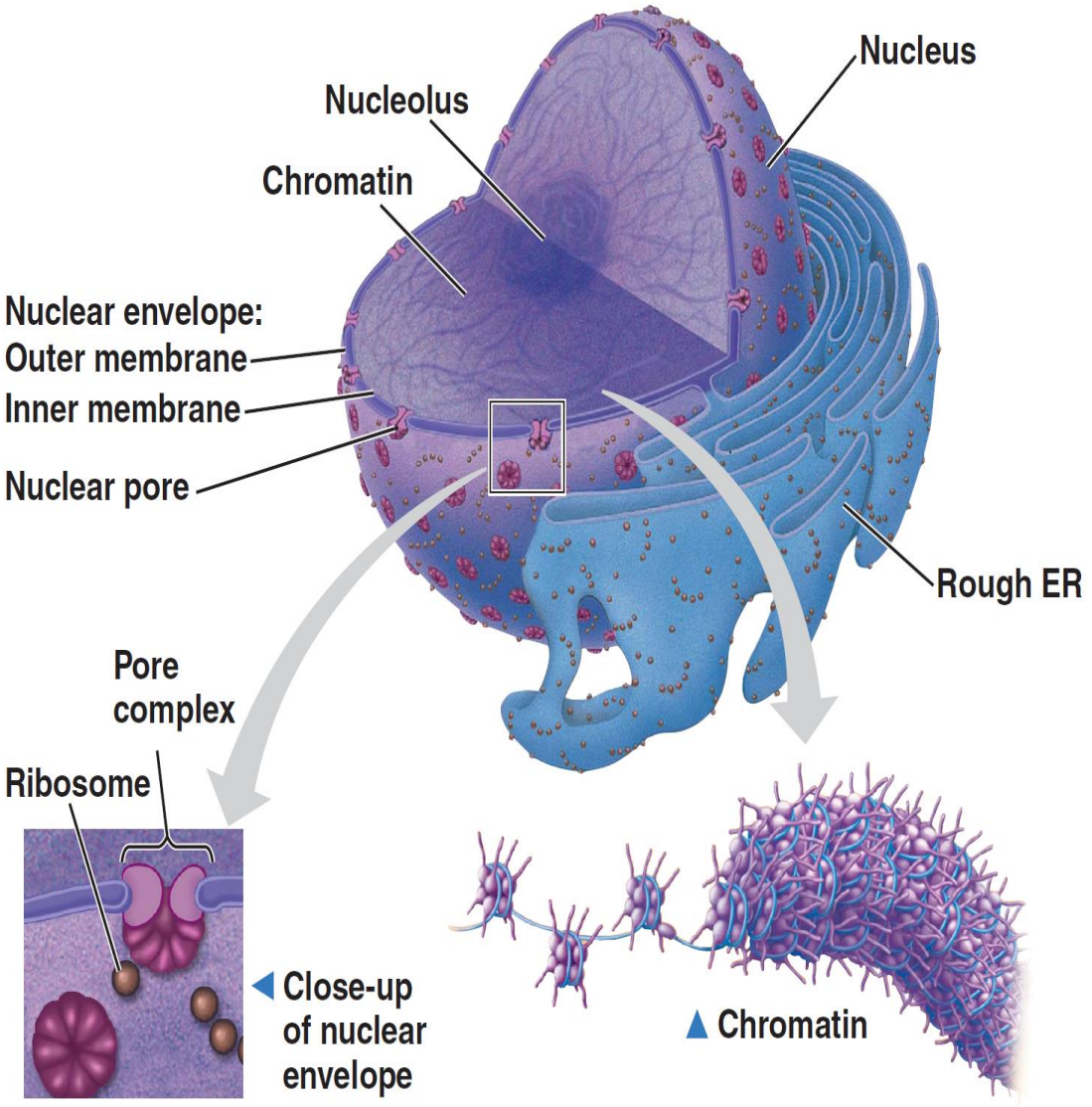
- The nucleus is separated from the cytoplasm by a *double* membrane called the **nuclear envelope**.

  - ✓ *The two membranes of the nuclear envelope are separated by 20–40 nm.*

  - ✓ *The **envelope is perforated by pores** that are about 100 nm in diameter.*

  - ✓ *At the lip of each pore, the inner and outer membranes of the nuclear envelope are fused to form a continuous membrane.*

  - ✓ *A protein structure called a **pore complex** lines each pore, regulating the passage of certain large macromolecules and particles.*

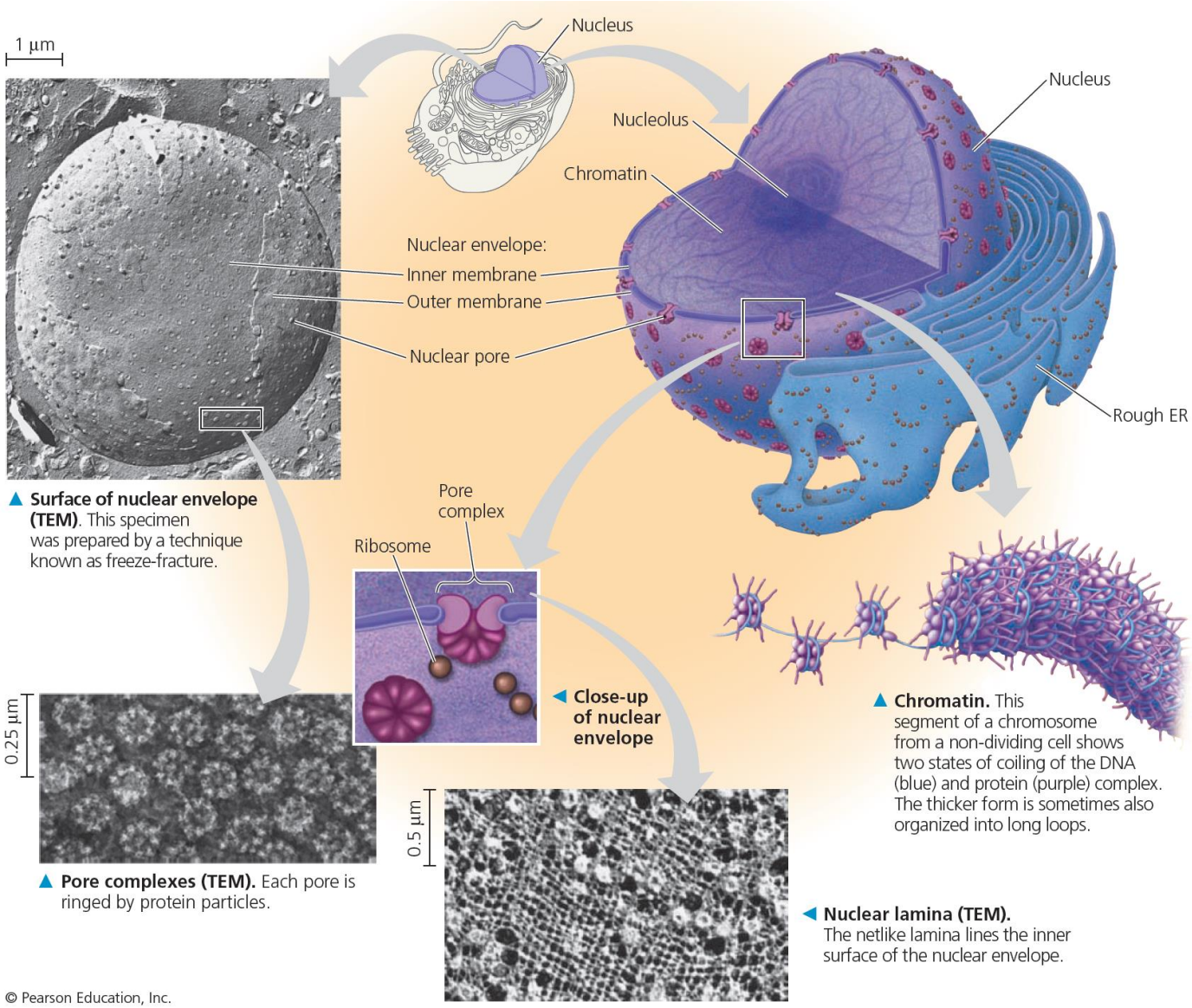


Nucleus and DNA

- The nuclear side of the envelope is lined by the **nuclear lamina**, a network of protein filaments that maintains the shape of the nucleus.

- Within the nucleus, the DNA and associated proteins are organized into discrete units called **chromosomes**, structures that carry the genetic information.

  - Each chromosome contains one long DNA molecule associated with many proteins. This complex of DNA and protein is called **chromatin**.



▲ **Surface of nuclear envelope (TEM).** This specimen was prepared by a technique known as freeze-fracture.

◀ **Close-up of nuclear envelope**

▲ **Chromatin.** This segment of a chromosome from a non-dividing cell shows two states of coiling of the DNA (blue) and protein (purple) complex. The thicker form is sometimes also organized into long loops.

▲ **Pore complexes (TEM).** Each pore is ringed by protein particles.

◀ **Nuclear lamina (TEM).** The netlike lamina lines the inner surface of the nuclear envelope.

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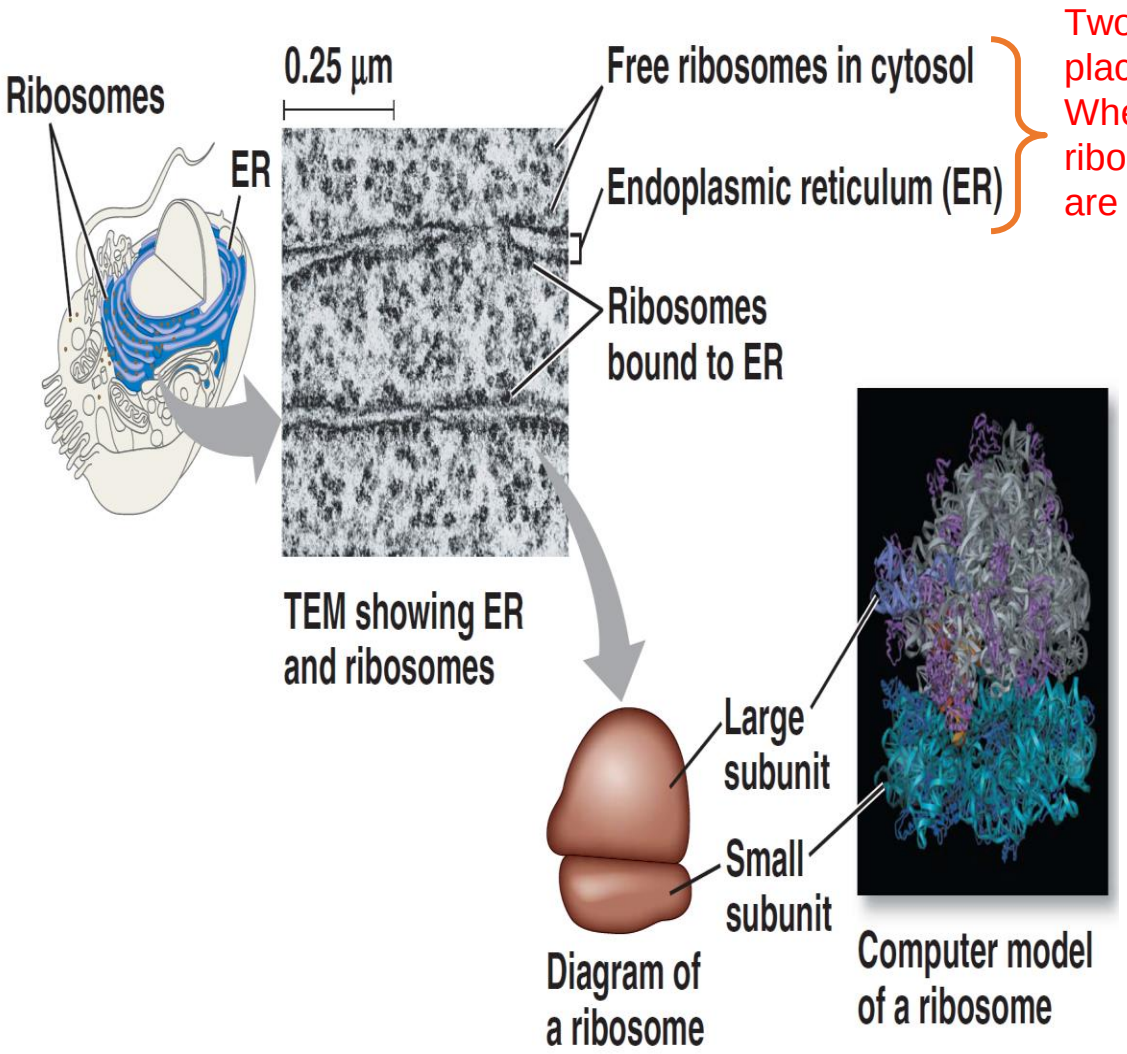
Nucleus and ribosomes

- In the nucleus is a region of densely stained fibers and granules adjoining chromatin, the **nucleolus**.
- **In the nucleolus, *ribosomal RNA* (rRNA) is synthesized** and assembled with proteins from the cytoplasm to form large and small ribosomal subunits.
  - *The subunits pass through the nuclear pores to the cytoplasm, where they combine to form ribosomes.*
- The nucleus directs protein synthesis by synthesizing messenger RNA (mRNA).
- The **mRNA is transported to the cytoplasm through the nuclear pores**.
- Once in the cytoplasm, ribosomes translate mRNA's genetic message into the primary structure of a specific polypeptide.



Two places Where ribosomes are found.

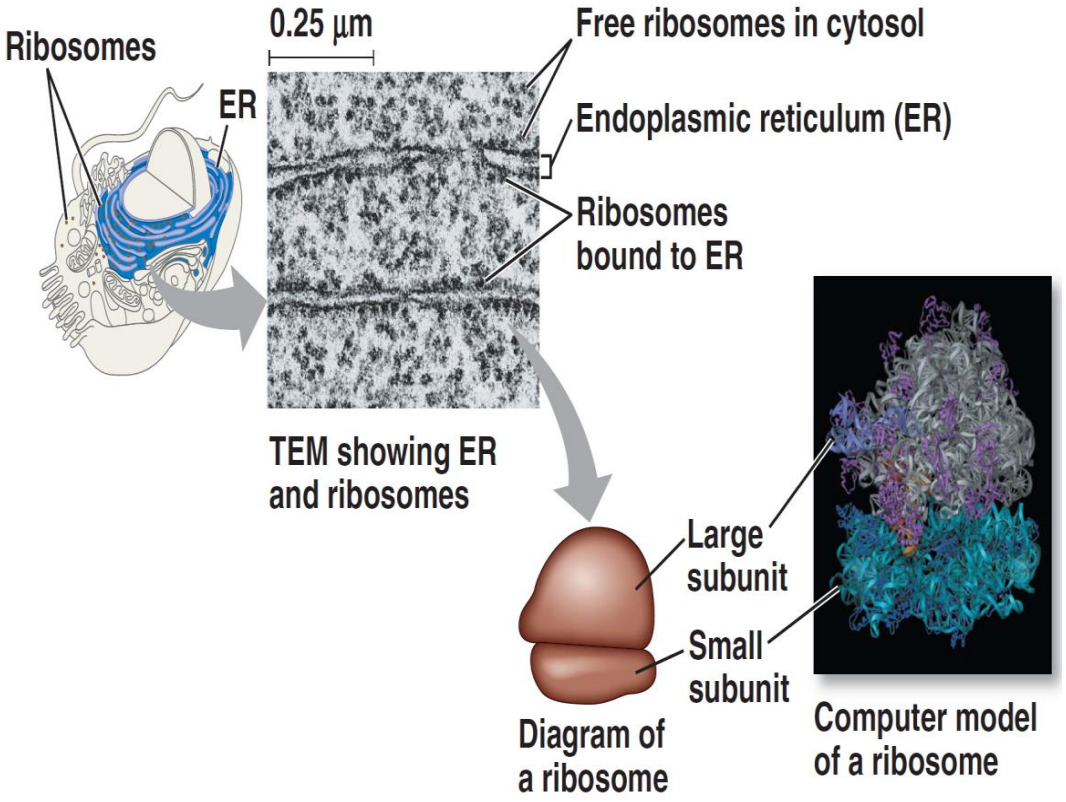
Ribosomes and protein factories

- **Ribosomes**, containing rRNA and protein, are the cellular components that carry out protein synthesis.
  - ✓ *Cell types that synthesize large quantities of proteins (such as pancreas cells) have large numbers of ribosomes and prominent nucleoli.*

- *Free ribosomes* are suspended in the cytosol and synthesize proteins that function within the cytosol.

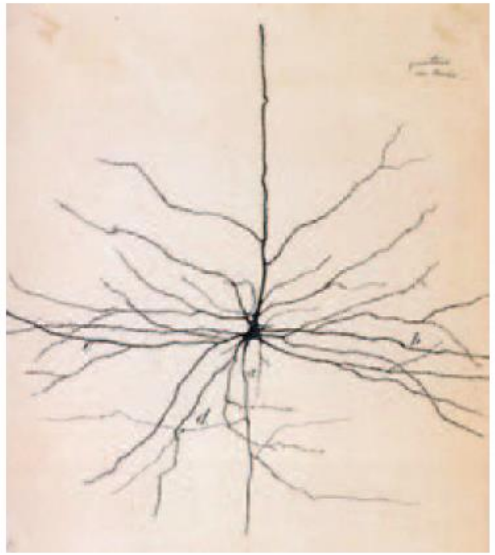
- *Bound ribosomes* are attached to the outside of the endoplasmic reticulum or nuclear envelope.

  - ✓ *Bound ribosomes synthesize proteins that are inserted into membranes, packaged into organelles such as ribosomes, or exported (secreted) from the cell.*
  - ✓ *Cells that specialize in protein secretion—for instance, the cells of the pancreas that secrete digestive enzymes—frequently have a high proportion of bound ribosomes.*



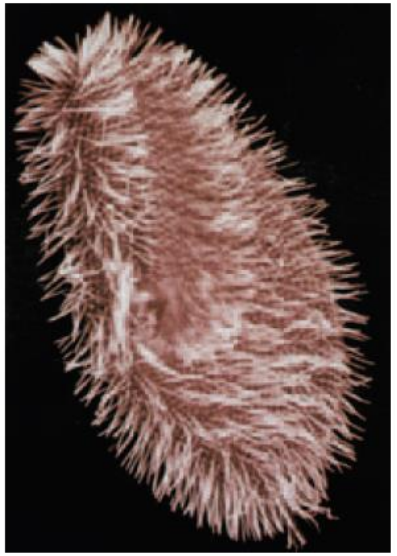
Extra slides

Unity in cells? I see diversity in size and shape.



(A) 100 µm

Neuron

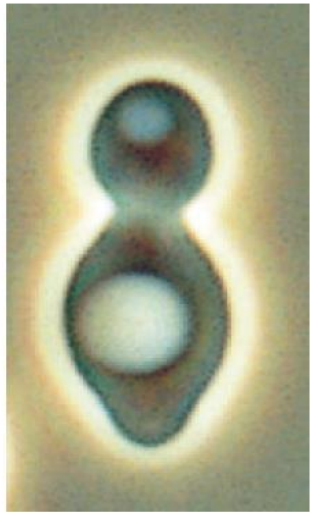


(B) 25 µm

Paramecium Protozoa

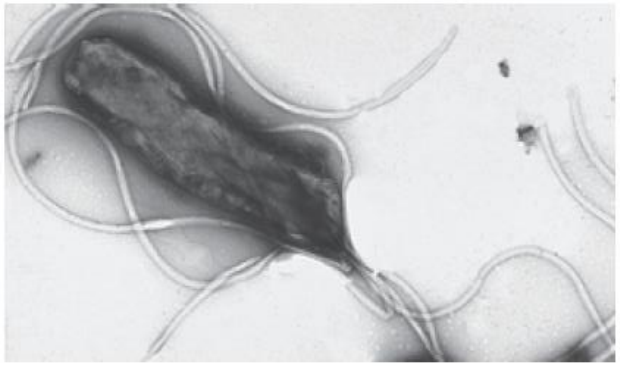
(C) 10 µm

Chlamydomonas.
single-celled green algae



(D) 5 µm

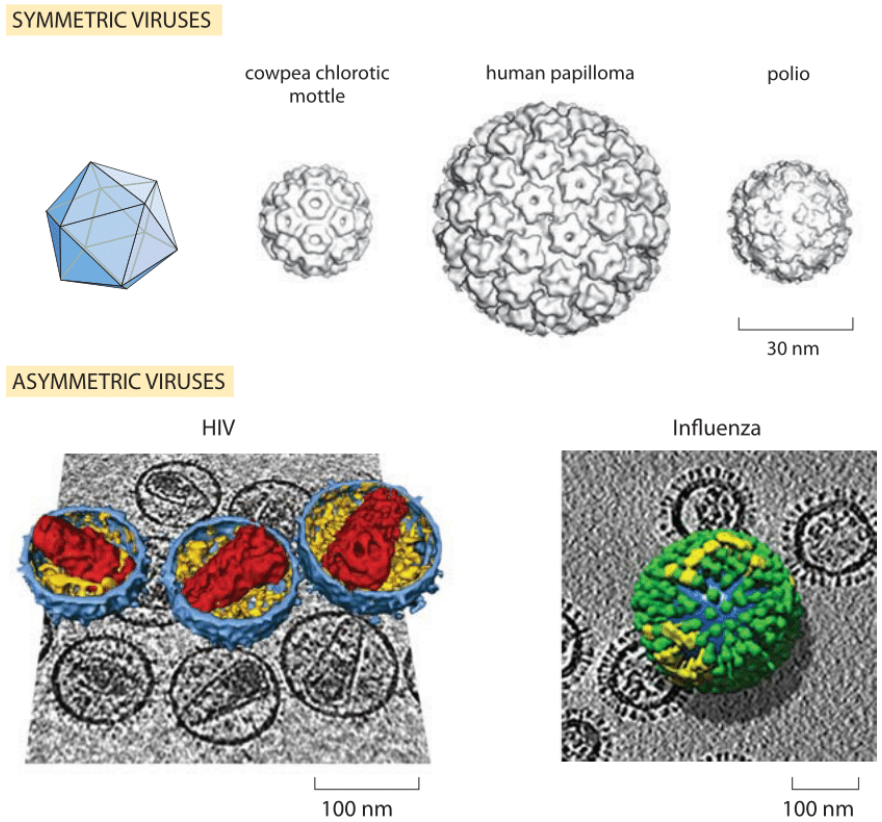
Saccharomyces cerevisiae.
This budding yeast cell.

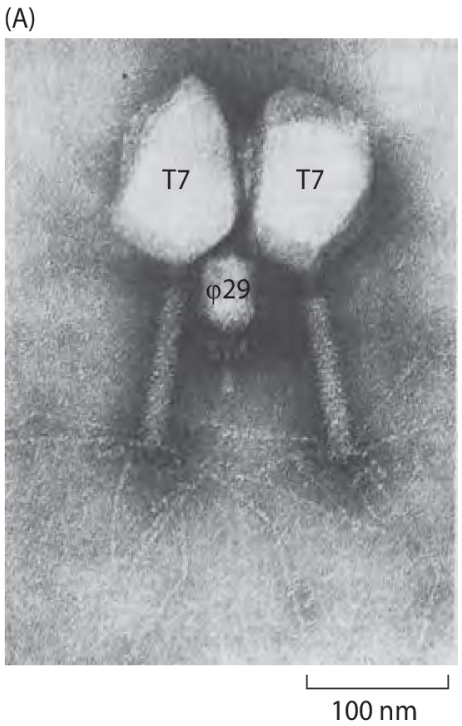


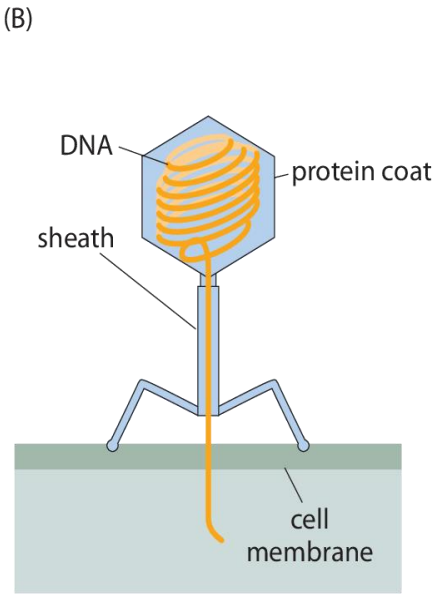
(E) 1 µm

Helicobacter pylori.

Shape and size of virus



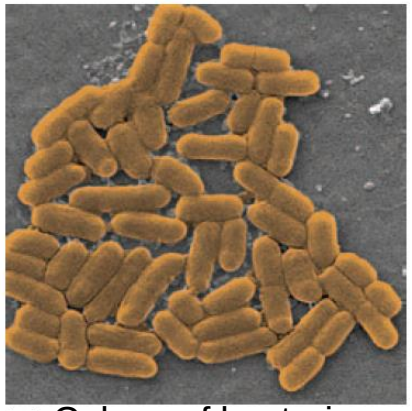




http://book.bionumbers.org/how-big-are-viruses/

http://book.bionumbers.org/ (SARS- virus information booklet with numbers)

Diversity? I see unity-- through biochemistry



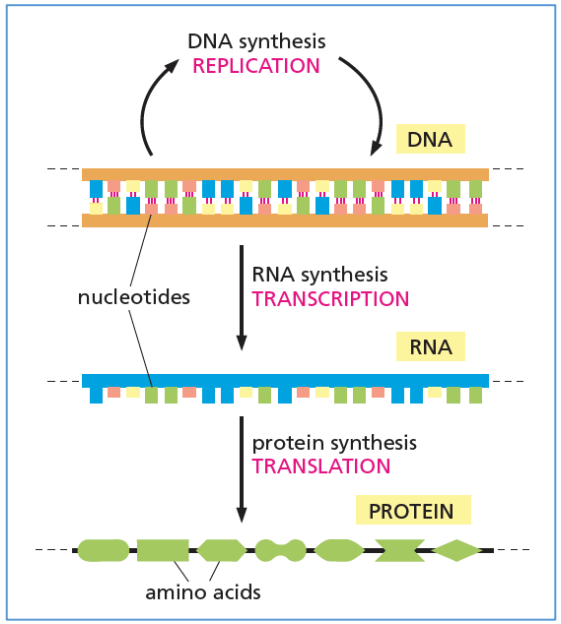
(A) Colony of bacteria

(B) Butterfly

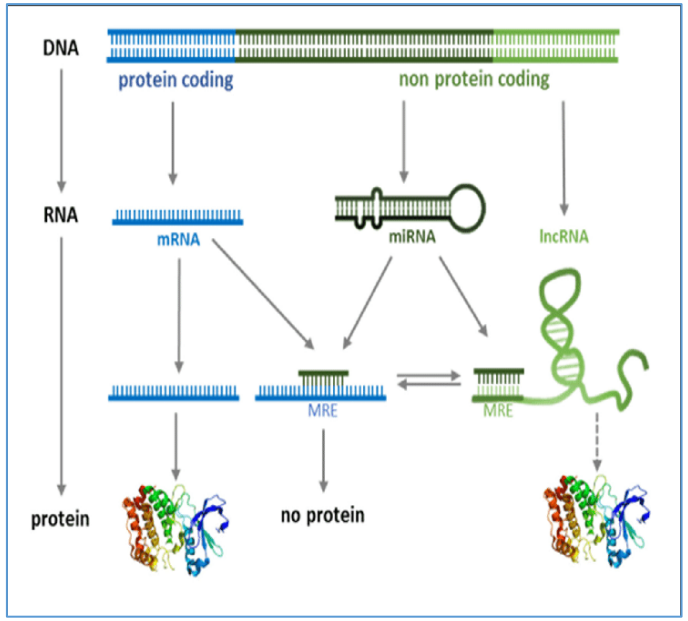
(C) Rose

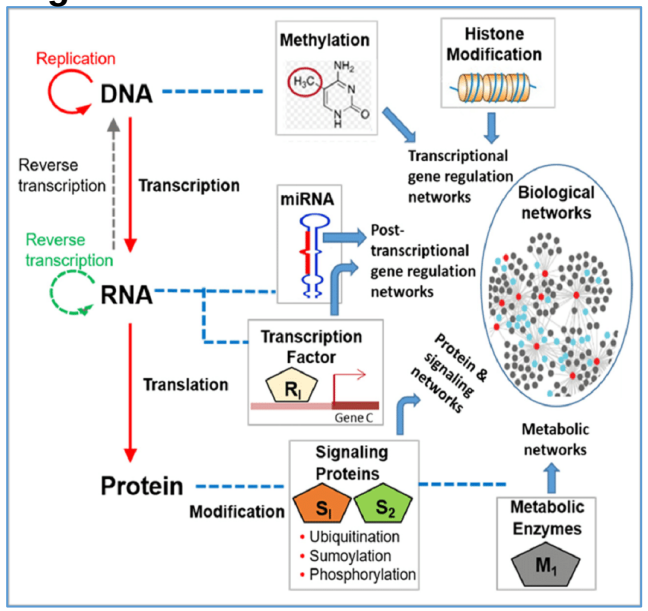
(D) Dolphin

Central dogma

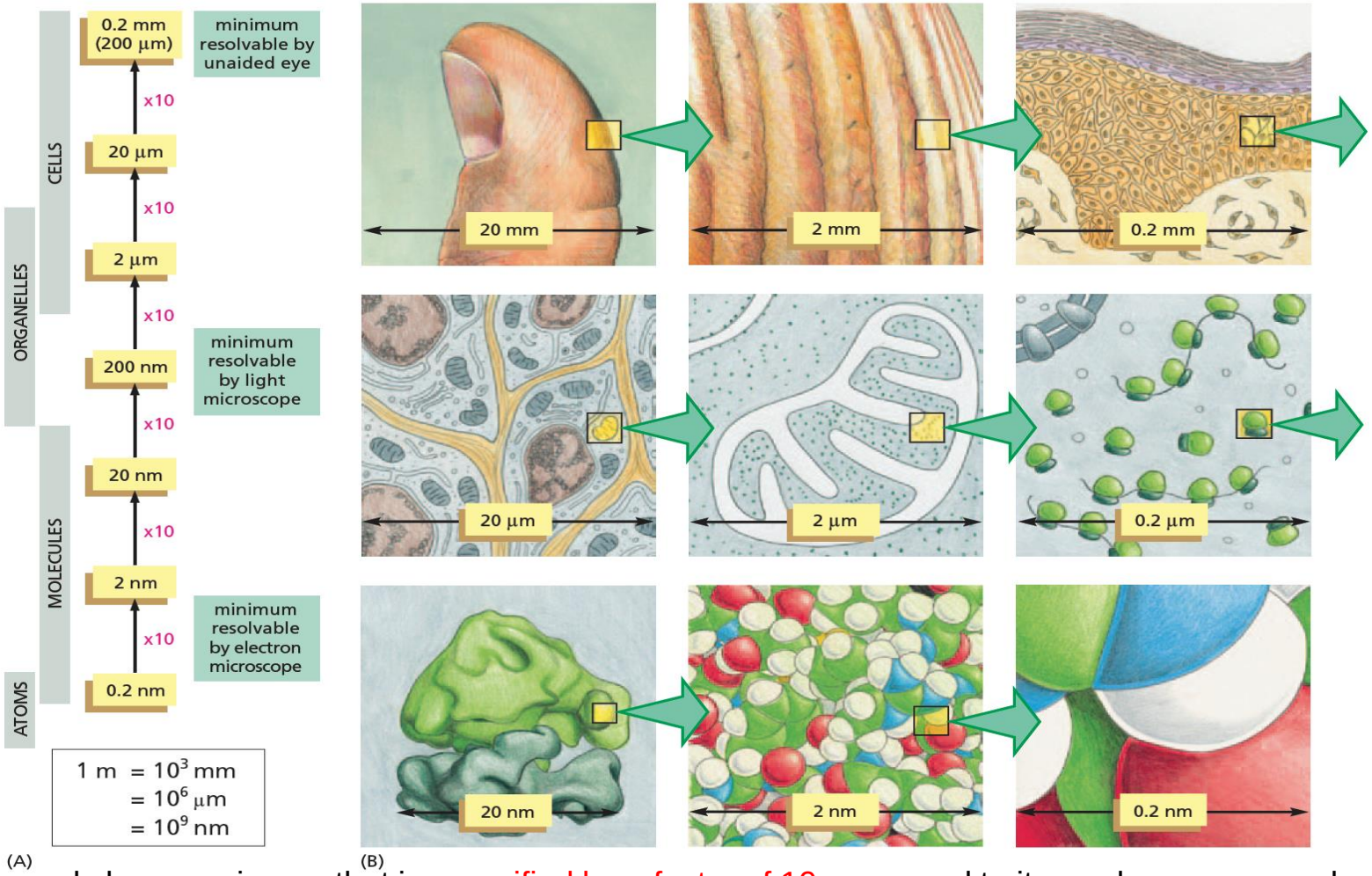


Modified central dogma





Cell size kaleidoscope -- Cell biology to biochemistry to biophysics to model?



Each panel shows an image that is magnified by a factor of 10 compared to its predecessor—producing an imaginary progression from a thumb, to skin, to skin cells, to a mitochondrion, to a ribosome, and ultimately to a cluster of atoms forming part of one of the many protein molecules in our bodies